

Seroefficacy of a 10-valent pneumococcal conjugate vaccine (Pneumosil®) compared with other pneumococcal conjugate vaccines in children

Oluwasefunmi Akeju^a, Shuo Feng^{a,b}, Ed Clarke^c, Adedapo O Bashorun^c, Ikechukwu Adigweme^c, Nicola Pidduck^d, Merryn Voysey^{a,e}

^aOxford Vaccine Group, Department of Paediatrics, University of Oxford, United Kingdom

^bSchool of Public Health, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

^cVaccines and Immunity Theme, MRC Unit The Gambia at the London School of Hygiene and Tropical Medicine, Fajara, The Gambia

^dNuffield Department of Primary Care Health Sciences, University of Oxford, Oxford, UK

^eNIHR Oxford Biomedical Research Centre, Oxford, United Kingdom

merryn.voysey@paediatrics.ox.ac.uk

Abstract

Background

A new 10-valent pneumococcal conjugate vaccine (PCV10^{SII}) was prequalified by WHO in 2019 for children based on non-inferiority of immunogenicity to PCV13 and PCV10^{GSK}. However, the efficacy of PCV10^{SII} against pneumococcal carriage or disease has not been compared with other PCVs. We compared the risk of seroinfection in PCV10^{SII} with PCV10^{GSK} and PCV13-vaccinated infants.

Methods

Data were available for 716 infants from two randomised trials comparing the immunogenicity of PCV10^{SII} with PCV13 and PCV10^{GSK}. Serotype-specific IgG levels were measured from serum samples collected at four weeks after the primary vaccination series and at the time of booster vaccination. We defined seroinfection as any increase in serotype-specific IgG between post-primary vaccination and the booster dose, indicating likely colonisation, subclinical infection, or disease. We compared the relative risk (RR) of seroinfection between PCVs.

Findings

There was no difference in the risk of seroinfection between PCV10^{SII} and PCV10^{GSK} for the shared serotypes, but PCV10^{SII} showed better protection against 6A and 19A, not included in PCV10^{GSK}. Compared with PCV13, the risk of seroinfection was 65% lower in PCV10^{SII} for

serotype 14 (RR= 0.35, 95% CI: 0.22, 0.57), 2.27-fold (95% CI: 1.40, 3.68) and 2.91-fold (95% CI: 1.84, 4.60) higher for 6A and 6B seroinfection, and similar for the remaining serotypes.

Interpretation

PCV10^{SII} likely offers similar protection as PCV10^{GSK} against colonisation by serotypes common to both and may offer better coverage in countries with high prevalence of 6A and 19A disease. The serotype-specific differences in seroefficacy between PCV10^{SII} and PCV13 should be considered when evaluating cost-effectiveness.

Funding

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Research in context

Evidence before this study

A new ten-valent pneumococcal conjugate vaccine (PCV10^{SII}) was prequalified by the World Health Organization in 2019 for use in children based on non-inferiority of its immunogenicity to PCV10^{GSK} and PCV13. We searched PubMed using terms that included “PCV10”, “Pneumosil”, “PCV10SII”, “PCV10-SII”, and “SIPL-PCV” on April 28, 2026, and found two studies that evaluated its effect on carriage, and both studies showed that PCV10^{SII} decreased vaccine-type (VT) carriage compared with unvaccinated children. However, no study has compared the effect of PCV10^{SII} on VT-carriage with other PCVs. This lack of data on comparative efficacy has been a limitation for models predicting the cost-effectiveness of switching from other PCVs to PCV10^{SII}. A previous published meta-analysis estimated seroinfection (a marker of colonisation, subclinical infection, or disease) from antibody data and compared the serotype-specific differences in seroinfection between older PCVs, an approach that we adopted in this study.

Added value of this study

This is the first study to compare the efficacy of PCV10^{SII} with other PCVs. We assessed the serotype-specific differences in seroinfection between PCV10^{SII} compared with PCV10^{GSK} and PCV13. We showed that PCV10^{SII} had similar seroefficacy to PCV10^{GSK} for shared serotypes, while compared with PCV13, PCV10^{SII} had higher seroefficacy against serotype 14 but lower seroefficacy against 6A and 6B. Overall, PCV13 showed better efficacy against any PCV10^{SII}-type seroinfection and seroinfection by multiple serotypes, in the study population. This analysis included two pairs of cross-reactive serotypes, 6A/6B and 19A/19F, which had not been assessed together in previous seroefficacy analyses. We adjusted for potential cross-reactivity when estimating the relative risks of seroinfection for each of these serotypes, an approach that will be useful for evaluating the seroefficacy of next-generation PCVs with increasing pairs of cross-reactive serotypes included.

Implications of all the available evidence

PCV10^{SII} and PCV10^{GSK} likely have similar protection against colonisation by serotypes common to both. There are serotype-specific differences in seroefficacy between PCV10^{SII} and PCV13 which should be considered, in addition to the local serotype distribution, when predicting impact or cost-effectiveness in a different population.

Introduction

Streptococcus pneumoniae remains a leading cause of pneumonia, invasive disease and mortality in children aged less than 5 years of age, causing more than 200,000 deaths per year, the majority of which occur in low- and middle-income countries (LMIC).¹ Since the seven-valent pneumococcal conjugate vaccine (PCV7) was first licensed in 2000, several other PCVs with increasing valency have become available. Most PCVs are expensive, and would not be affordable to low or middle income countries (LMIC)s without support from Gavi, the Vaccine Alliance.² The first two PCVs that became available through Gavi were the 13-valent (PCV13, Prevenar 13®) and 10-valent PCV (PCV10^{GSK}, Synflorix®). Both PCV10^{GSK} and PCV13 have shown high effectiveness against vaccine-type (VT) invasive pneumococcal disease (IPD).³ However, there is need for sustainably affordable PCVs now that 17 countries have been transitioned out of Gavi support to self-financing for PCV, and the transition for several others is imminent.⁴ Moreso, as of 2023, more than 20 LMICs, ineligible for Gavi support, are yet to introduce PCVs.² A novel 10-valent PCV (PCV10^{SI}, Pneumosil®) was developed by the Serum Institute of India as a more affordable product for the LMIC market.⁵ It was the third PCV to become available through Gavi, being offered at \$2 per dose, which is at least 27% lower than the Gavi price for PCV10^{GSK} or PCV13.⁶

PCV10^{SI} includes capsular polysaccharides for serotypes 1, 5, 6A, 6B, 7F, 9V, 14, 19A, 19F and 23F, individually conjugated to a recombinant non-toxic diphtheria cross-reactive material 197 (CRM₁₉₇) protein.⁷ All the 10 serotypes in PCV10^{SI} are included in PCV13, which has an additional three serotypes (3, 4 and 18C). PCV10^{SI} has eight serotypes in common with PCV10^{GSK}, but serotypes 4 and 18C in PCV10^{GSK} are replaced by 6A and 19A in PCV10^{SI}.⁷ PCV10^{SI} was prequalified by WHO in 2019, based on non-inferiority of immune responses compared with PCV10^{GSK} and PCV13,⁸ and two studies conducted in India and Niger have investigated and shown the effectiveness of PCV10^{SI} against VT carriage compared with unvaccinated children.^{9,10} To the best of our knowledge, there has been no published evidence comparing its efficacy against disease or carriage with other PCVs. Some LMICs, including India and South Africa have switched from other PCVs to PCV10^{SI} mostly because of its affordability. Besides affordability, having some evidence comparing the efficacy of PCV10^{SI} to current PCVs would help other countries in their policy decision-making regarding PCV switching, especially LMICs that have limited capacity for conducting disease surveillance and effectiveness studies after introducing or switching to a new vaccine.

Previously, immunogenicity data have been used to evaluate vaccine 'seroefficacy'—the efficacy against seroinfection.¹¹ Seroinfection, an immunological indicator of subclinical infection,

carriage or disease, is defined as an increase in antibody levels between the post-primary vaccination timepoint and the booster dose—a period when rapid waning is expected in infants.¹¹ Experimental human challenge models of *S. pneumoniae* have shown that carriage, but not mere exposure (i.e., challenge without carriage), results in increased anti-capsular (serotype-specific) antibody levels in the serum,¹² although intranasal challenge without carriage can transiently boost local antibodies in the nasal mucosa.¹³ Exceptions to this serotype specificity of antibody response to *S. pneumoniae* infection or immunisation have been observed in some pairs of serotypes such as 6A and 6B, and 19A and 19F, which have demonstrated cross-reactivity due to their structural similarity.¹⁴ Seroinfection allows for head-to-head comparisons of efficacy across different PCVs without the resource-intensive process of frequent swabbing in carriage studies. A meta-analysis comparing the immunogenicity of PCV7, PCV10^{GSK} and PCV13 showed that higher post-vaccination antibody levels correlated with better protection against seroinfection, regardless of differences in the PCV formulations.¹¹ Similarly, in this study, using immunogenicity data from two clinical trials, we compared the serotype-specific IgG levels and the seroefficacy of PCV10^{SII} with PCV13 and PCV10^{GSK}.

Methods

Study design

This study is a secondary analysis of antibody data collected from two randomised clinical trials comparing the immunogenicity of PCV10^{SII} with PCV13 or PCV10^{GSK}.

Data sources

We searched public databases including Cochrane Central Register of Controlled Trials, Clinicaltrials.gov, Embase and Medline, for randomised trials that compared the immunogenicity of PCV10^{SII} with PCV13 or PCV10^{GSK} in infants and reported serotype-specific IgG estimates at 4-6 weeks after the primary vaccination series (at the peak of vaccine-induced IgG) and another before the booster vaccination.^{5,7} We found and included two eligible studies and obtained individual participant level data from the authors and data owners of the studies.

Description of included studies

The two studies, A (NCT03896477) and B (NCT02308540), were conducted in the Western region of The Gambia from 2019-2020 and 2015-2016 respectively. In study A, 660 infants aged 6-8 weeks were randomised 1:1:1 to receive either PCV10^{SII}, PCV13 or PCV10^{GSK} in a 2+1 schedule at 6 and 14 weeks, followed by a booster dose at 9-18 months. Study B included 200 infants aged 6-8 weeks randomised 1:1 to receive either PCV10^{SII} or PCV13 in a 3+1 schedule

with primary doses at 6, 10 and 14 weeks, and a booster dose at 10-14 months. Of the 660 enrolled infants in study A, 620 (208 PCV10^{SII}, 206 PCV13 and 206 PCV10^{GSK}) received a booster dose. In study B, 96 of the 200 infants (49 PCV10^{SII}, 47 PCV13) received a booster dose. Each vaccine dose was 0.5ml and their exact components, including carrier proteins, have been previously described.⁷ Serum samples were obtained at four weeks after the third primary dose, on the day of booster vaccination, and four weeks after the booster vaccination. Serum IgG against each of the 10 PCV10^{SII} serotypes were estimated at the WHO pneumococcal reference laboratory at University College London, using validated enzyme-linked immunosorbent assay (ELISA). Additionally, both studies collected safety data on solicited and unsolicited adverse events throughout the follow-up period, including events unrelated to the vaccine.

Outcomes

The primary outcome was seroinfection by any of the serotypes included in PCV10^{SII} vaccine. The geometric mean ratio (GMR) of serotype-specific IgG between vaccine arms at one month after the primary series was the intermediate outcome.

Statistical analysis

Immunogenicity

The 620 participants from study A and the 96 participants from study B with pre-booster IgG data were included in the analysis. IgG levels below the limit of quantification (LOQ), which was 0.15 mg/ml, were assigned a value of one-half the LOQ (0.075 mg/ml). There were a few participants whose post-primary or post-booster IgG measurement was missing for one or more serotypes, and these were excluded only when estimating the GMR for the serotype with missing data, since their seroinfection status for that serotype cannot be assessed. We report the serotype-specific GMR of post-primary IgG between vaccine groups (PCV10^{SII} versus PCV13 in both studies, PCV10^{SII} versus PCV10^{GSK}, and PCV10^{GSK} versus PCV13 arms in study A only), for the eligible participants included in each analysis. Our study cohort for each analysis was smaller than the published cohorts in which the authors estimated the GMR and therefore results differ slightly. We calculated the pooled serotype-specific GMR for post-primary IgG in PCV10^{SII} versus PCV13 arms in both studies, using fixed-effects meta-analysis.

Seroinfection

Seroinfection, a binary variable, was assigned a value of 1 with any numerical increase in serotype-specific IgG from post-primary vaccination to pre-booster sampling timepoint, and 0 if otherwise. We calculated the proportion of participants with seroinfection in each vaccine group and the relative risk (RR) of seroinfection for each serotype; any of the 10 PCV10^{SII}

serotypes; any of the eight serotypes shared with PCV10^{GSK}; and seroinfection with multiple serotypes; between the vaccine groups. RR was not computed when no seroinfection occurred in any of the two comparison groups. We used fixed-effects meta-analysis to calculate the pooled RR of seroinfection between PCV10^{SII} and PCV13 arms in both studies but reported RR from only one study when no RR was calculated in the other study, due to absence of seroinfection events. Relative seroefficacy was calculated as one minus RR of seroinfection.¹¹ To account for potential cross-reactivity between 6A and 6B, we adjusted for 6B seroinfection when estimating the RR of 6A seroinfection across vaccine groups in a binary regression model, and vice versa. Additionally, we conducted sensitivity analysis by calculating RR of either 6A or 6B seroinfection after excluding cases of co-seroinfection with both 6A and 6B, based on the assumption that these might be due to cross-reactivity rather than true co-infection. The same approaches were used to account for potential cross-reactivity between 19A and 19F.

Association between immunogenicity and seroefficacy

To evaluate the association between immunogenicity and seroefficacy across all serotypes, we combined the post-primary GMR and RR for each serotype and the four pairs of PCV comparisons from both studies (PCV10^{SII} versus PCV13 in both studies, PCV10^{SII} versus PCV10^{GSK}, and PCV10^{GSK} versus PCV13), generating up to 40 RR-GMR pairs of values. We regressed the log₂(RR) of seroinfection on log₂(GMR), using linear regression and assessed the Pearson's correlation between both variables. We did not investigate the association between RR and pre- or post-booster IgG because the pre-booster IgG levels were often too low and confounded by seroinfection, while post-booster IgG would have no effect on seroinfection in the period prior to receiving the booster.

Role of the funding source

The funder of the study had no role in data collection, data analysis, data interpretation, or writing of the report.

Results

Serotype specific GMR after primary dose

The pooled GMRs comparing post-primary vaccination IgG in PCV10^{SII} to PCV13 favoured PCV13 over PCV10^{SII} for six serotypes (5, 6A, 7F, 9V, 19A and 19F), while GMRs favoured PCV10^{SII} over PCV13 for serotype 14. There was no difference in GMRs for the remaining serotypes (Figure 1a). In the comparison of PCV10^{SII} with PCV10^{GSK}, except 19F where GMR favoured PCV10^{GSK}, GMRs were 1.35 to 3.78-fold higher in PCV10^{SII} for the other seven shared

serotypes (1, 5, 6B, 7F, 9V, 14 and 23F). As expected, GMRs were also higher in PCV10^{SII} for 6A and 19A which are not included in PCV10^{GSK} (Figure 1b).

Prevalence of seroinfection

The prevalence of seroinfection by any of the 10 PCV10^{SII} serotypes was 57% (355/620) in study A and 50% (48/96) in study B. Among cases in studies A and B, seroinfection was caused by only one serotype in 49% (175/355) and 58% (28/48) of children; two serotypes 34% (121/355) and 25% (12/48) of children; and ≥ 3 serotypes in 17% (59/355) and 17% (8/48) of children, respectively. The most prevalent serotypes in seroinfection were 6A, 6B and 19A among PCV10^{GSK} and PCV10^{SII} vaccinated infants. Among PCV13 vaccinated infants, serotype 14 was the most prevalent seroinfection in both studies, and 6A, 6B, 19A, and 19F were also relatively common in this group (Figure 2). Serotypes 1, 5 and 9V were the least prevalent across all the study arms in both studies. Across both studies, PCV13 vaccinated infants had a lower prevalence of seroinfection from multiple serotypes (based on the 10 shared serotypes) compared with PCV10^{SII} (34% vs 51%; $p = 0.006$). There was no difference in the prevalence of seroinfection by multiple serotypes between PCV10^{SII} and PCV10^{GSK} vaccinated infants, with all 6A and 19A cases excluded (22% vs 29%, $p = 0.30$) (Supplementary table 1).

The risk of seroinfection by any of the 10 shared serotypes was 26% higher in PCV10^{SII} arm compared with PCV13 (RR= 1.26; 95% CI= 1.05, 1.52). On the other hand, between PCV10^{SII} and PCV10^{GSK}, there was no difference in the risk of seroinfection from any of the eight shared serotypes common to both (RR= 1.00; 95% CI= 0.81, 1.24). (Figure 3).

Serotype-specific risk of seroinfection

The pooled serotype-specific risk of seroinfection was higher for PCV10^{SII} compared with PCV13 for serotypes 6A (RR= 2.27, 95% CI: 1.40, 3.68) and 6B (RR= 2.91, 95% CI: 1.84, 4.60), both in univariate analysis and after adjusting for potential cross-reactivity. Conversely, the risk of serotype 14 seroinfection was 65% lower in PCV10^{SII} compared with PCV13 (RR= 0.35, 95% CI: 0.22, 0.57). There was no difference in the relative risk of seroinfection between PCV10^{SII} and PCV13 for the other serotypes. In the comparison between PCV10^{SII} and PCV10^{GSK}, there was no difference in the risk of seroinfection for any of the shared serotypes. (Figure 3, Supplementary figures 2-3).

Cross-reactive serotypes

Across both studies, 40% (287/709) of infants had either 6A or 6B seroinfection and 28% (81/287) of these were positive for both 6A and 6B. Similarly, 18% (124/699) of infants had

either 19A or 19F seroinfection, and 10% (12/124) of these were positive for both (Supplementary tables 2-3). The risk of 6A seroinfection between vaccine groups was similar after adjusting for 6B seroinfection; likewise, the risk of 19A after adjusting for 19F seroinfection, and vice versa. Also, the RRs of 6A, 6B, 19A and 19F seroinfection between vaccine groups remained similar when co-seroinfection cases were excluded (Supplementary tables 4-5).

Association between immunogenicity and seroefficacy

Across all serotypes, the log-GMR of IgG at post-primary vaccination was negatively correlated with the log-RR of seroinfection (Pearson's $r = -0.68$, $p < 0.001$) (Figure 4). The fitted linear model showed that PCVs that induced similar antibody response (GMR of 1) achieved similar seroefficacy (RR= 1.06; 95% CI: 0.82, 1.36). And a two-fold increase in antibody response (GMR= 2) was associated with a 35% reduction in the risk of seroinfection (RR= 0.65; 95% CI: 0.54, 0.79).

Discussion

This is the first study to compare the seroefficacy of PCV10^{SII} with other licensed vaccines. PCV10^{SII} and PCV10^{GSK} showed similar protection against seroinfection for the eight serotypes common to both. Additionally, PCV10^{SII} offered better protection against 6A and 19A, which are not in PCV10^{GSK} highlighting the importance of understanding local serotype prevalence when deciding between PCV products. We observed serotype-specific differences in seroefficacy between PCV10^{SII} and PCV13 for three shared serotypes. Compared with PCV13, PCV10^{SII} showed better protection against serotype 14 while PCV13 was better at protecting against 6A and 6B. Overall, the risk of seroinfection by any of the 10 serotypes was 26% higher in PCV10^{SII} than PCV13 in this study cohort. Some models that previously evaluated the cost-effectiveness of switching from other PCVs to PCV10^{SII} were limited by lack of data on the serotype-specific comparative efficacy of PCV10^{SII}.^{15,16} The serotype-specific differences in seroefficacy from this study can provide more reliable estimates for future cost-effectiveness models.

We observed a negative correlation between the immunogenicity (GMR) of PCV and the risk of seroinfection (RR) for all serotypes, in line with that observed by Feng et al in a large meta-analysis of global studies that compared the seroefficacy of PCV13 to PCV10^{GSK}.¹¹ Like Feng, we found that PCVs with the same immunogenicity provided similar protection: Our model predicted a 35% (RR= 0.65 95% CI: 0.54, 0.79) decrease in the risk of seroinfection for every two-fold rise in antibody levels, which is within the confidence limits of the 54% (RR= 0.46; 95% CI: 0.23, 0.96) decreased risk estimated previously.¹¹ These further strengthen the validity

of our findings and suggest that the seroefficacy estimates are likely to be relevant to other settings.

Additionally, our seroefficacy analyses included two pairs of cross-reactive serotypes (6A/6B and 19A/19F), whereas only one from each pair, 6B and 19F, were previously assessed.¹¹ Hence, we adjusted for potential cross-reactivity when comparing the serotype-specific risk of seroinfection across vaccine groups although our findings suggest that the influence of cross-reactivity on the validity of the adjusted estimates of risks for these serotypes might be negligible.

Seroinfection was detected in at least half of the children in the included studies. Most seroinfections likely represented nasopharyngeal carriage or other subclinical infections.^{5,7} Carriage is a pre-requisite for invasive pneumococcal disease (IPD) and pneumococcal transmission.¹⁷ Therefore, a vaccine with higher protection against seroinfection resulting from reduced carriage acquisition will be better at preventing IPD and providing indirect protection by reducing transmission. Furthermore, as pneumococcal carriage is associated with increased risk of viral RTI,¹⁸ and PCVs can reduce the risk of some viral RTIs, reducing carriage can have impacts beyond those due to *S. pneumoniae*.^{19,20}

The Gambia introduced PCV7 in 2009 but switched to PCV13 in 2011, using a 3+0 schedule and achieving a coverage of >90% for all three doses from 2011-2018 and ≥75% from 2019.²¹ The high prevalence of vaccine-type (VT) seroinfection (10 serotypes) in 2019-2020 (study A) even among PCV13-vaccinated infants (40%) suggests continued circulation of some vaccine serotypes after eight years of PCV13 introduction. The estimated VT carriage (all PCV13 serotypes), measured by swabbing, among children 0-4 years from three post-PCV surveys (2015-2022) in The Gambia was 10% to 17.5%.^{22,23} The higher prevalence of VT seroinfection compared with VT carriage is unsurprising as seroinfection is a measure of multiple carriage events that have occurred over a longer period of time, many of which would have cleared by the time of swabbing. Seroinfection could therefore be more sensitive for detecting transient or cleared carriage events which can be missed during single cross-sectional or even repeated periodic swabbing.

However, some seroinfections represent antibody cross-reactivity rather than true infections. Additionally, seroinfection might not detect some cases of carriage or disease occurring shortly after the primary vaccination series as any boost in antibody levels that has waned and returned to baseline levels before pre-booster sample collection would be missed. An over- or under-

estimation of the true number of carriage events due to cross-reactivity or any lack of sensitivity will likely move any between-group comparisons towards the null as these add 'noise' to any statistical comparisons of randomized groups, rather than bias.

In conclusion, we have compared the efficacy of PCV10^{SII} with PCV13 and PCV10^{GSK}. PCV10^{SII} and PCV10^{GSK} offer similar protection against seroinfection for the eight shared serotypes, after the primary vaccination doses in infants. PCV10^{SII} might offer better overall protection than PCV10^{GSK} in countries where 6A and 19A are more prevalent. PCV10^{SII} had lower seroefficacy than PCV13 for serotypes 6A and 6B and better seroefficacy against serotype 14. Future comparisons of cost-effectiveness between these PCVs should account for the local serotype prevalence and the serotype-specific differences in seroefficacy.

Contributors

MV and SF conceived and designed the study, and obtained funding and participant data. NP and OA devised and conducted the search strategy. OA, SF, and MV contributed to the methods, accessed, verified, and analysed the data. OA wrote the first draft of the manuscript. EC, AOB, and IA led the clinical trials that generated the data for the study. All authors interpreted the data and contributed to the writing of the final version of the manuscript. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Data Sharing

Data were accessed by applying for use from the data owners for both studies. A data sharing agreement was signed to allow use of the data solely for the purposes of the research and does not allow further onward sharing of the data.

Declaration of interests

Acknowledgements

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Figures

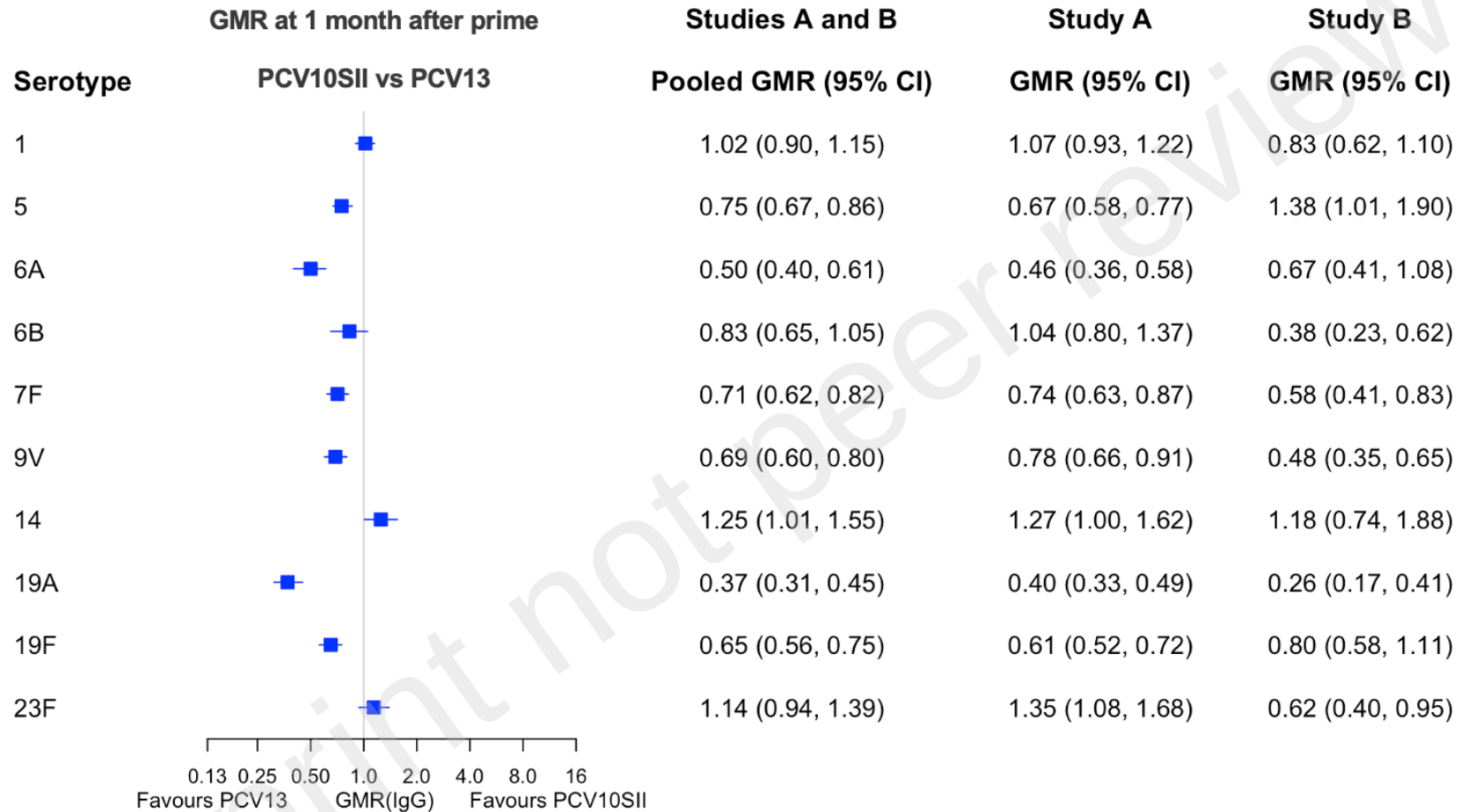


Figure 1a. Geometric mean ratios (GMR) of serotype-specific IgG between PCV10^{SII} vs PCV13 at one month after the primary vaccination series. The forest plot shows the pooled GMR and the confidence intervals (represented by the blue boxes and lines) from both studies A and B, estimated using fixed-effects meta-analysis. And the two columns on the far right shows the GMR from each study.

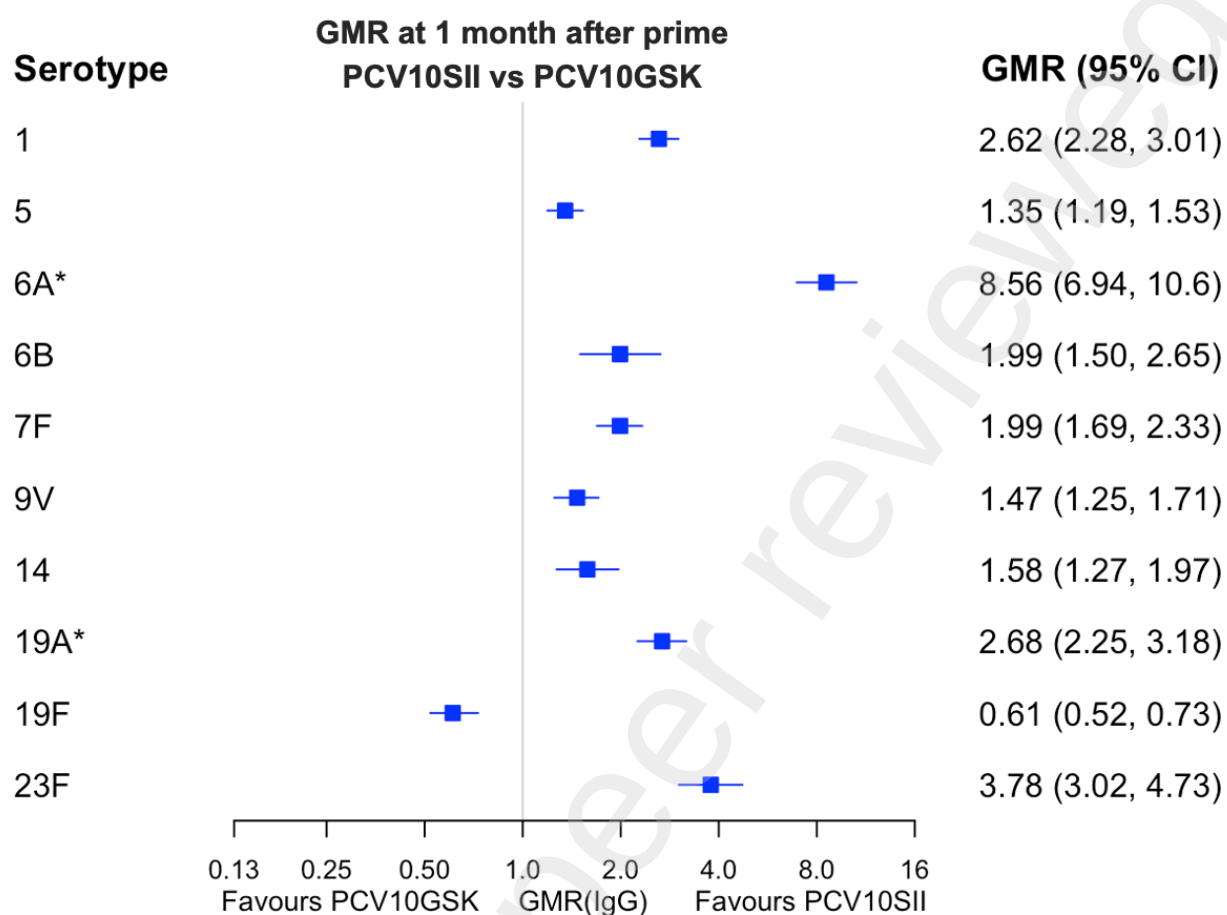


Figure 1b. Geometric mean ratios (GMR) of serotype-specific IgG between PCV10^{SII} vs PCV10^{GSK} at one month after the primary vaccination series. The blue boxes and lines of the plot represent the GMR and confidence intervals.

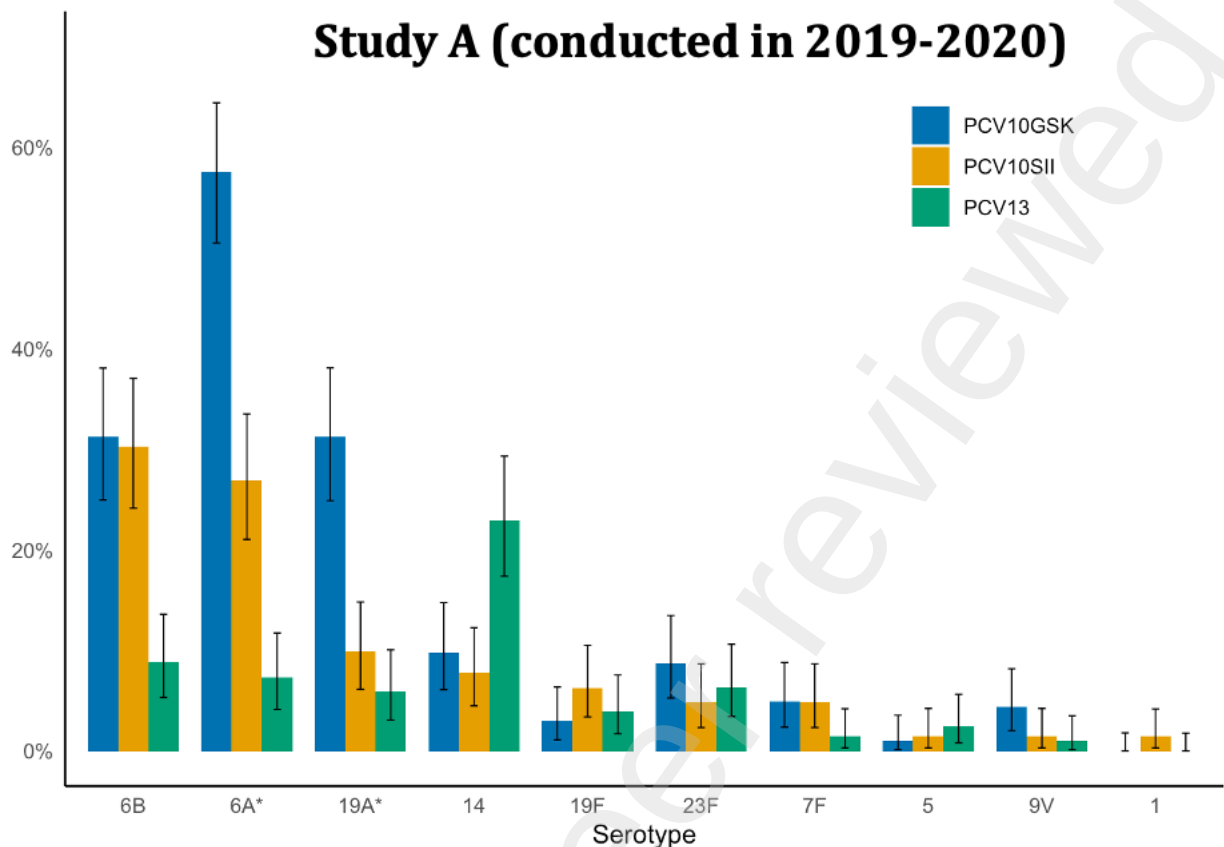


Figure 2a. Prevalence of seroinfection by PCV10^{SII} serotypes among PCV10^{GSK}, PCV10^{SII} and PCV13-vaccinated infants in study A, conducted in 2019-2020. The bars represent the percentage of seroinfection caused by each of the PCV10^{SII} serotypes while the lines indicate the confidence intervals.

*6A and 19A are not included in PCV10^{GSK}.

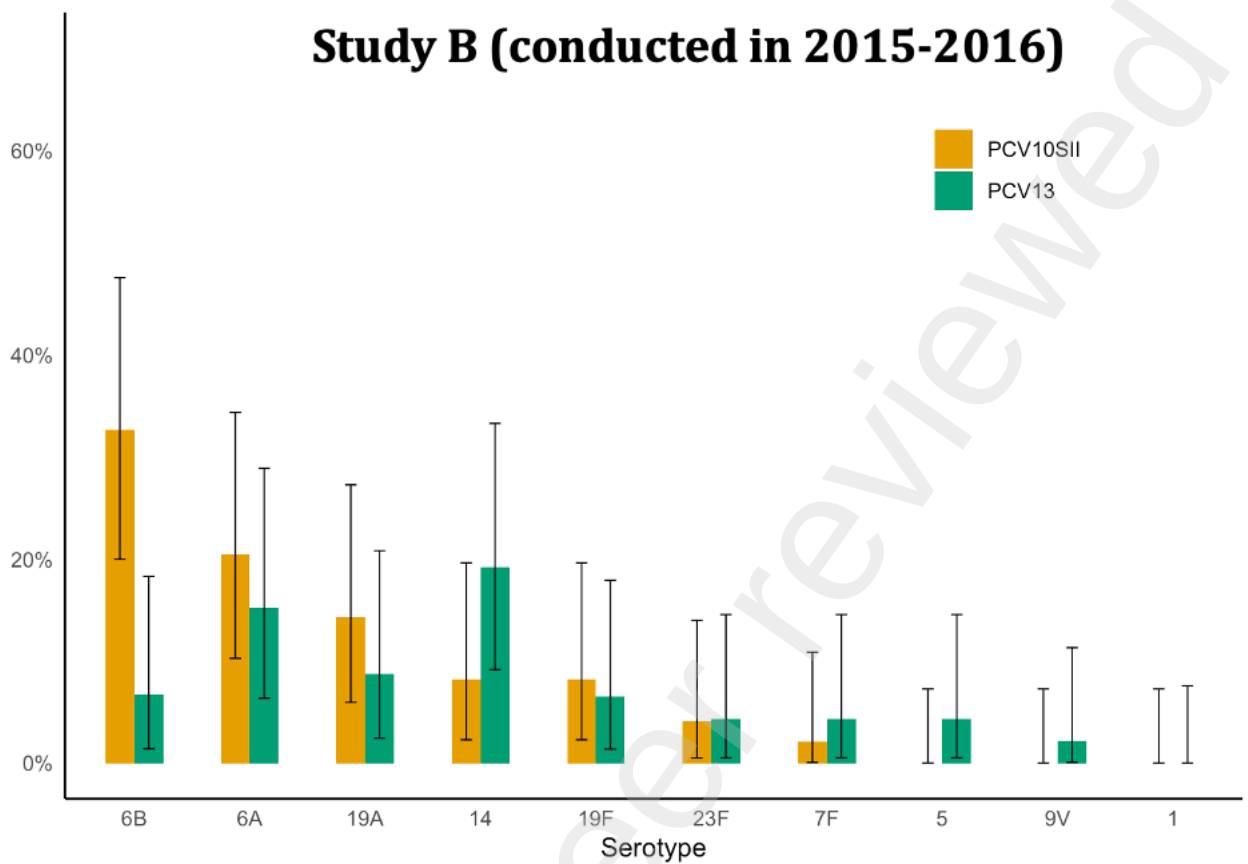


Figure 2b. Prevalence of seroinfection by PCV10^{SII} serotypes among PCV10^{SII} and PCV13-vaccinated infants in Study B, conducted in 2015-2016. The bars represent the percentage of seroinfection caused by each of the PCV10^{SII} serotypes while the lines indicate the confidence intervals.

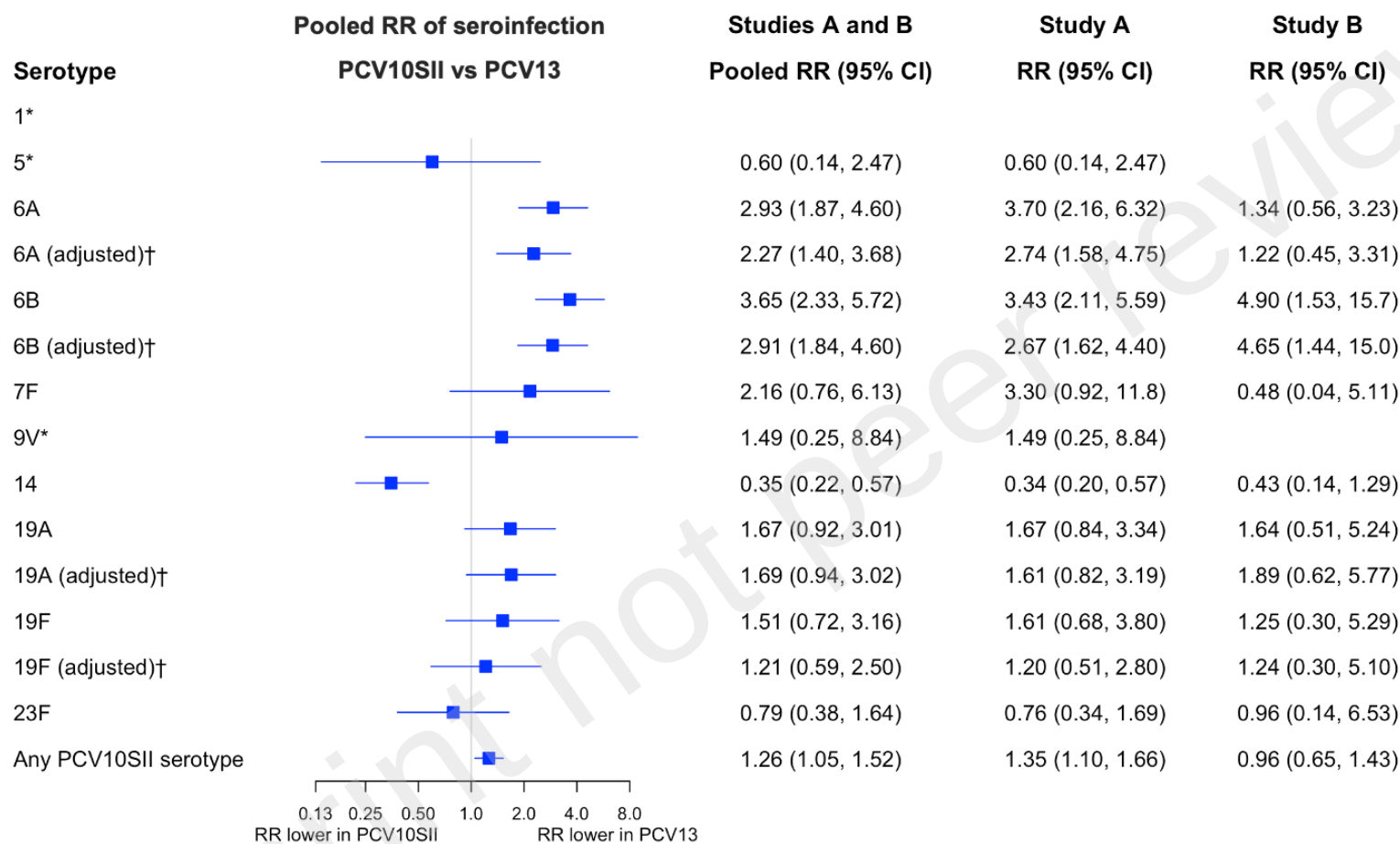


Figure 3a. Serotype-specific relative risks (RR) of seroinfection between PCV10^{SII} vs PCV13 (ref: PCV13). The forest plot shows the pooled RR and the confidence intervals (represented by the blue boxes and lines) from studies A and B, estimated using fixed-effects meta-analysis. And the two columns on the far right shows the RR from each study. *RR for serotype 1 was not calculated because of zero case in at least one of the arms in each study, and the pooled RR for 5 and 9V were from study A only because of zero case in one of the arms in study B. †RR of 6A and 6B seroinfection were adjusted for 6B and 6A seroinfection; and RR of 19A and 19F seroinfection, adjusted for 19F and 19A seroinfection, respectively.

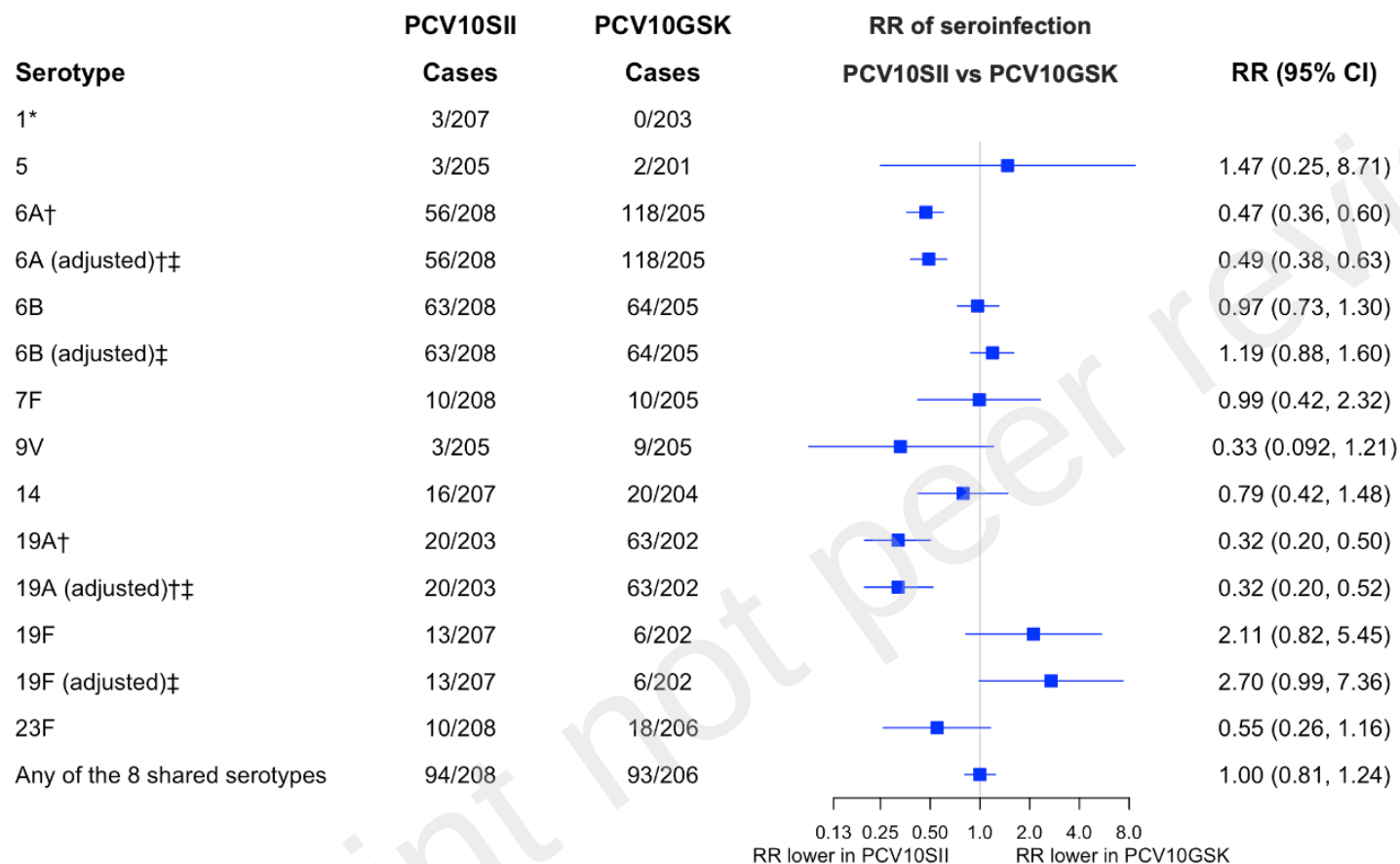


Figure 3b. Serotype-specific relative risks (RR) of seroinfection between PCV10^{SII} vs PCV10^{GSK} (ref: PCV10^{GSK}). The blue boxes and lines of the plot represent the RR and confidence intervals. *RR for serotype 1 was not calculated because of zero case in one of the arms. † 6A and 19A are not included in PCV10^{GSK}. ‡ RR of 6A and 6B seroinfection were adjusted for 6B and 6A seroinfection; and RR of 19A and 19F seroinfection, adjusted for 19F and 19A seroinfection, respectively. The columns to the left of the plot describes the seroincidence (proportion of cases) for each serotype among PCV10^{SII} and PCV10^{GSK} vaccinated infants.

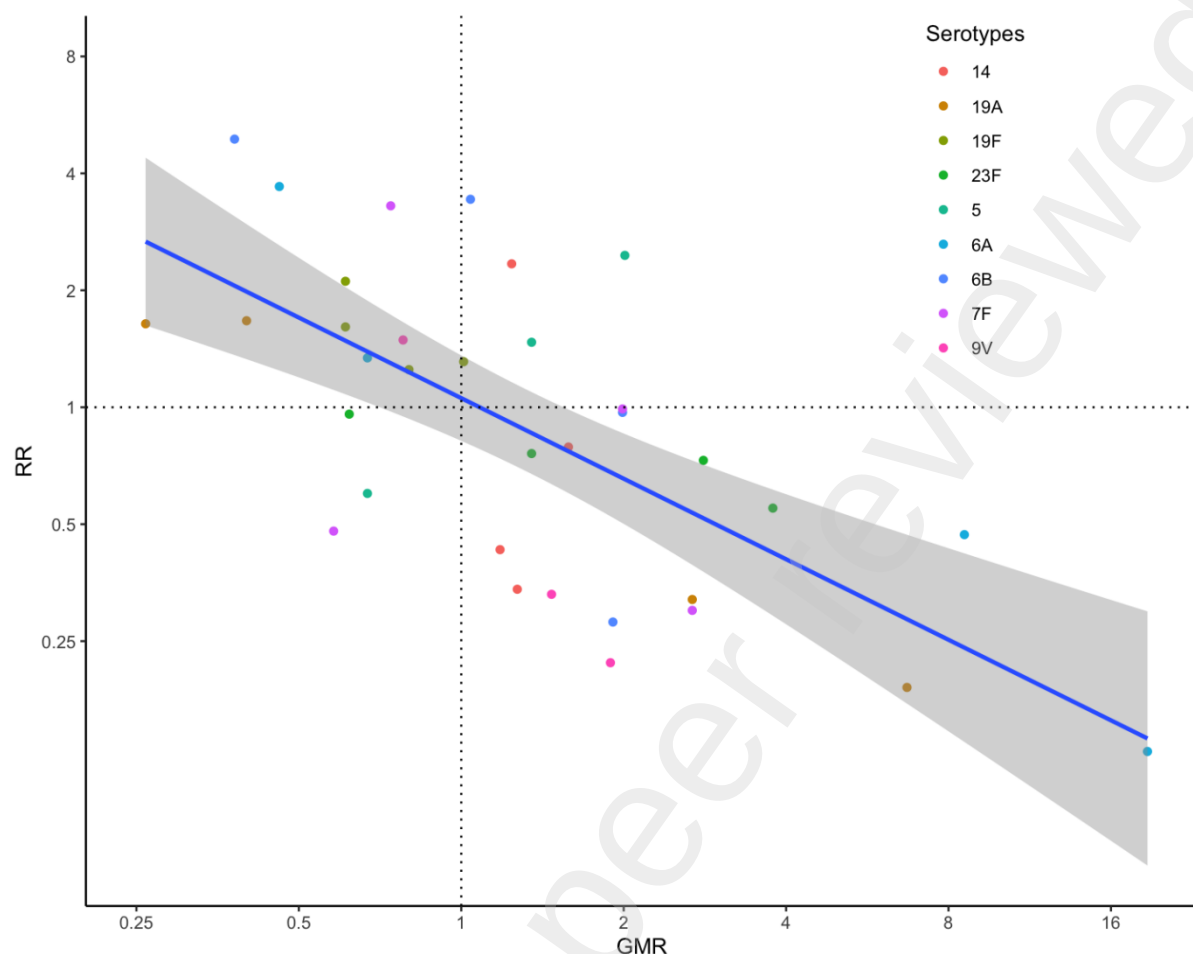


Figure 4a. Association between geometric mean ratio of IgG and the relative risk of seroinfection across PCV10^{SII} serotypes. Each point represents a serotype-specific comparison of GMR and RR between two vaccine groups in a study on a log₂ scale. The four pairs of vaccine groups compared are: PCV10^{SII} versus PCV13 (two pairs, each from studies A and B); PCV10^{SII} versus PCV10^{GSK}; and PCV13 versus PCV10^{GSK}. The dotted gridlines indicate the point on the x-axis where GMR= 1 and y-axis where RR= 1. The solid blue line shows the association between the predicted log₂(RR) and log₂(GMR) from the linear regression model, and the shaded area represents the confidence intervals. Pearson's correlation between log₂(GMR) and log₂(RR) was -68% (p<0.001). Serotype 1 was excluded because RR was not calculated due to zero case in at least one of the PCV arms in each study, and only the RR and GMR estimates from study A were included for 5 and 9V because of zero case in one of the arms in study B.

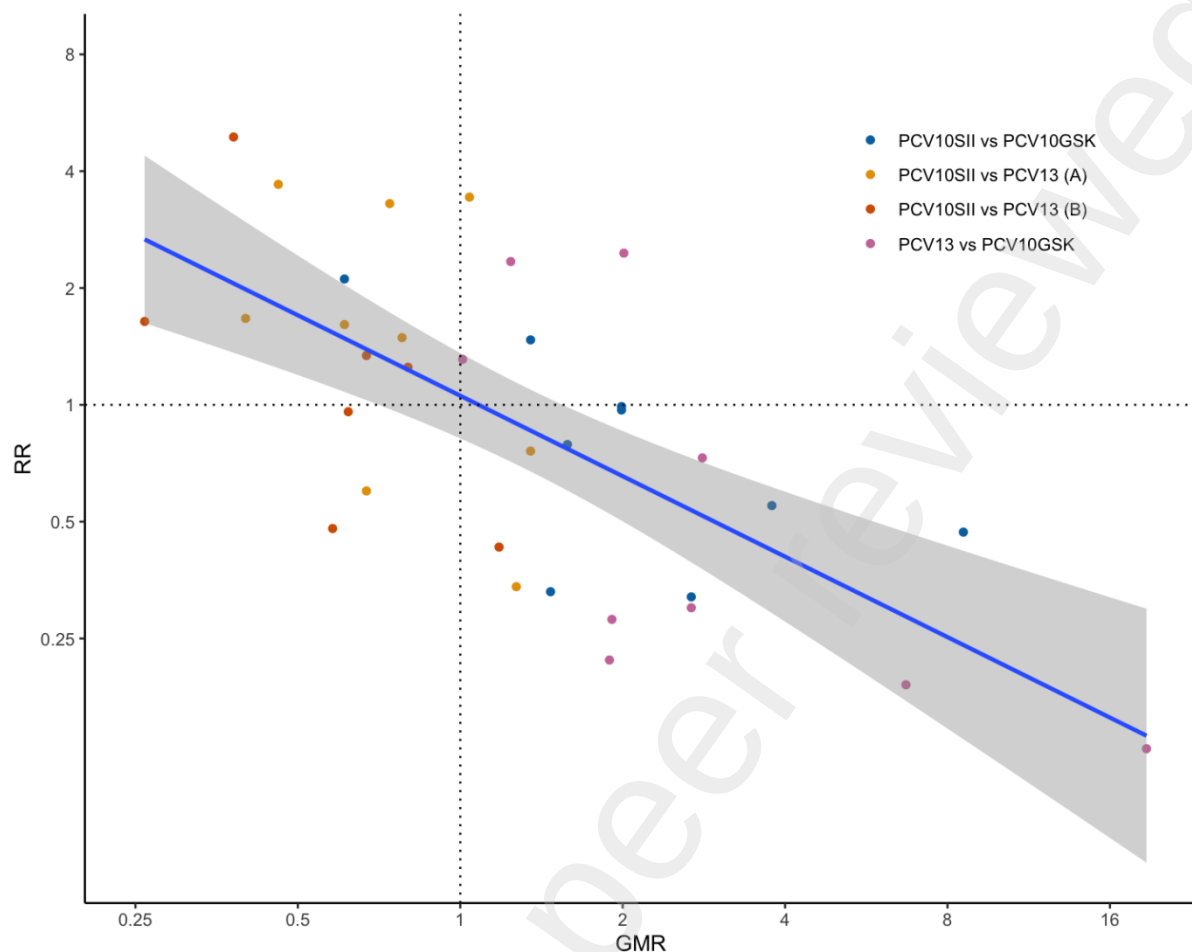


Figure 4b. Association between geometric mean ratio of IgG and the relative risk of seroinfection across PCV10^{SII} serotypes (same plot as Figure 4a but classified by vaccine comparison groups). Each point represents a serotype-specific comparison of GMR and RR between two vaccine groups in a study, on a log₂ scale. The four pairs of vaccine groups compared are: PCV10^{SII} versus PCV13 (two pairs, each from studies A and B); PCV10^{SII} versus PCV10^{GSK}; and PCV13 versus PCV10^{GSK}. The dotted gridlines indicate the point on the x-axis where GMR= 1 and y-axis where RR= 1. The solid blue line shows the association between the predicted log₂(RR) and log₂(GMR) from the linear regression model, and the shaded area represents the confidence intervals. Pearson's correlation between log₂(GMR) and log₂(RR) was -68% (p<0.001). Serotype 1 was excluded because RR was not calculated due to zero case in at least one of the PCV arms in each study, and only the RR and GMR estimates from study A were included for 5 and 9V because of zero case in one of the arms in study B.